

TORONTO BUTTONVILLE, ON, Canada

WMO: 716390

Lat: 43.8660N

Lon: 79.3681W

Elev: 650

StdP: 14.35

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.9	1.6	-15.0	2.5	-2.2	-9.0	3.5	2.7	24.1	21.2	21.9	21.3	6.6	340	0.545

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	88.9	72.1	85.6	70.2	82.6	68.8	74.4	84.8	72.5	82.1	70.9	79.5	10.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.2	117.7	79.2	69.4	110.8	77.6	67.8	104.5	76.3	38.3	85.0	36.6	82.4	35.2	79.3	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.0	18.6	16.7	-10.6	93.9	7.0	3.4	-15.7	96.4	-19.8	98.4	-23.7	100.3	-28.8	102.7
			-11.4	77.6	6.8	2.3	-16.3	79.3	-20.3	80.6	-24.1	81.9	-29.1	83.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.8	21.5	23.0	31.8	43.8	56.0	66.1	70.8	69.2	62.0	49.7	38.3	28.4
	DBStd	19.32	11.73	10.26	10.34	8.37	8.22	6.80	5.64	5.43	7.10	7.69	8.22	9.33
	HDD50	3595	884	755	571	220	31	0	0	0	2	102	359	671
	HDD65	7141	1349	1175	1030	636	301	66	11	21	138	476	802	1136
	CDD50	2445	0	0	6	34	218	483	646	594	364	93	7	0
	CDD65	516	0	0	0	0	23	99	192	151	49	2	0	0
	CDH74	4366	0	0	2	15	257	884	1669	1114	402	23	0	0
	CDH80	1269	0	0	0	1	63	281	531	289	101	3	0	0
Wind	WSAvg	7.5	8.4	8.4	8.6	8.7	7.6	6.9	6.6	6.1	6.3	7.0	7.6	8.0
Precipitation	PrecAvg	31.7	2.4	1.9	2.2	2.8	3.0	3.0	3.3	2.9	2.6	2.6	2.8	2.4
	PrecMax	38.5	5.3	3.8	4.5	5.0	5.6	7.4	5.8	5.8	6.1	5.3	5.3	4.4
	PrecMin	22.9	1.1	0.7	0.9	0.9	0.9	0.4	0.6	0.7	0.5	0.8	1.1	0.9
	PrecStd	4.4	0.9	0.9	1.0	1.1	1.1	1.7	1.3	1.1	1.3	1.3	1.1	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.7	52.1	68.9	75.9	86.0	91.7	93.1	91.6	88.6	76.9	63.4	54.3
		MCWB	49.4	46.0	57.0	60.7	68.2	73.4	74.4	72.8	71.8	65.3	55.7	50.2
	2%	DB	44.7	44.6	58.2	68.7	81.0	87.2	89.3	86.4	82.4	71.3	58.4	48.4
		MCWB	41.8	40.3	49.5	55.9	65.5	70.6	72.5	71.1	69.2	61.9	53.2	45.2
	5%	DB	40.1	40.1	51.5	62.9	75.8	82.9	85.7	82.9	78.6	66.5	54.6	44.0
		MCWB	37.7	36.6	44.6	51.7	62.5	68.8	70.5	69.4	67.0	59.0	49.8	41.1
	10%	DB	36.6	36.6	45.7	57.9	71.1	79.0	82.4	80.2	74.7	62.2	50.7	40.3
		MCWB	34.3	33.6	39.8	48.1	59.4	66.2	68.7	67.8	64.8	56.2	46.6	37.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.0	46.6	59.5	63.0	71.8	75.5	77.6	75.9	74.2	68.3	57.8	51.1
		MCDB	51.5	51.6	67.2	72.6	82.5	88.9	88.7	84.8	83.9	74.2	60.7	53.4
	2%	WB	42.3	41.1	51.2	58.3	67.6	72.5	74.7	73.4	71.4	63.4	54.1	45.9
		MCDB	44.0	43.9	56.4	66.0	78.1	83.9	85.2	83.1	79.7	69.4	57.9	47.9
	5%	WB	38.0	36.9	45.2	53.5	64.2	70.2	72.6	71.7	68.8	60.1	50.2	41.5
		MCDB	39.9	39.5	51.2	61.3	74.2	80.2	82.1	79.5	75.3	65.0	53.6	43.8
	10%	WB	34.5	33.9	40.6	49.3	60.8	68.0	70.8	69.9	66.6	57.5	47.0	37.7
		MCDB	36.4	36.3	45.1	56.2	68.8	76.0	79.5	77.6	73.0	61.6	50.4	40.0
Mean Daily Temperature Range	5% DB	MDBR	13.9	14.7	15.7	17.8	19.3	18.7	18.9	18.3	18.8	15.9	13.4	11.7
		MCDBR	14.8	17.4	24.0	27.1	25.3	23.9	22.7	21.8	23.7	22.7	19.5	14.9
		MCWBR	14.0	14.6	17.3	17.6	14.1	12.2	10.6	10.2	12.4	14.7	15.4	13.4
	5% WB	MCDBR	14.8	16.8	22.8	23.9	22.7	21.5	19.6	18.3	19.9	19.9	17.8	14.5
		MCWBR	14.4	15.1	17.5	17.7	14.0	12.1	10.2	9.7	11.8	14.3	15.4	14.0
Clear-Sky Solar Irradiance	taub	0.296	0.312	0.328	0.373	0.404	0.430	0.425	0.413	0.380	0.359	0.316	0.295	
	taud	2.375	2.328	2.403	2.319	2.282	2.236	2.269	2.292	2.370	2.411	2.487	2.431	
	Ebn at Noon	263	278	289	282	276	268	267	267	268	258	255	253	
	Edh at Noon	24	31	33	39	41	43	42	39	34	28	22	21	
All-Sky Solar Radiation	RadAvg	445	690	1101	1409	1708	1854	1868	1634	1304	776	481	349	
	RadStd	44	68	111	147	180	138	134	83	121	73	62	38	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page